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Course: BCH 402

Institution: Federal University, Birnin Kebbi

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Paper 1 of 1 — BCH 402

Level 400 • Second Semester • Handout

BCH 402: BIOSYNTHESIS OF MACROMOLECULES (1 UNIT)

OUTLINE TOPICS

Bacterial cell wall synthesis

Synthesis of complex lipids

Lipoproteins

Nucleic acids

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BACTERIAL CELL WALL SYNTHESIS

In order to retain shape and withstand intracellular pressure, most bacteria are surrounded by a cell wall that consists mainly of the cross-linked polymer peptidoglycan (PG).

The cell wall is the principal stress-bearing and shape-maintaining element in bacteria, and its integrity is of critical importance to cell viability. In both gram-positive and gram-negative bacteria, the scaffold of the cell wall consists of the cross-linked polymer peptidoglycan (PG).

Peptidoglycan or murein is a unique large macromolecule, a polysaccharide, consisting of sugars and amino acids that forms a mesh-like peptidoglycan layer outside the plasma membrane, the rigid cell wall characteristic of most bacteria.

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The sugar component consists of alternating residues of β -(1,4) linked N -acetylglucosamine (NAG) and N -acetylmuramic acid (NAM).

Attached to the N-acetylmuramic acid is an oligopeptide chain made of three to five amino acids.

Peptidoglycan serves a structural role in the bacterial cell wall, giving structural strength, as well as counteracting the osmotic pressure of the cytoplasm. This repetitive linking results in a dense peptidoglycan layer which is critical for maintaining cell form and withstanding high osmotic pressures, and it is regularly replaced by peptidoglycan production.

Peptidoglycan hydrolysis and synthesis are two processes that must occur in order for cells to grow and multiply, a technique carried out

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in three stages:

Bacterial Cell Wall Synthesis Continue.....

clipping of current material, insertion of new material, and re-

crosslinking of existing material to new material

The peptidoglycan layer is substantially thicker in Gram-positive bacteria (20 to 80 nanometers) than in Gram negative bacteria (7 to 8 nanometers). Depending on pH growth conditions, the peptidoglycan forms around 40 to 90% of the cell wall's dry weight of Gram-positive bacteria but only around 10% of Gram-negative strains.

It is difficult to tell whether an organism is gram-positive or gram-negative using a microscope; gram staining, created by Hans Christian Gram in 1884, is required. The bacteria are stained with several dyes such as crystal violet, iodine alcohol, and safranin using the gram staining procedure. Gram positive cells are purple after staining, while

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Gram negative cells are red.

Figure 1: Structure of Peptidoglycan

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Bacterial Cell Wall Synthesis Continue.....

The peptidoglycan layer in the bacterial cell wall is a crystal lattice structure formed from linear chains of two alternating amino sugars, namely N-acetylglucosamine (GlcNAc or NAG) and N-acetylmuramic acid (MurNAc or NAM).

The alternating sugars are connected by a β -(1,4)-glycosidic bond. Each MurNAc is attached to a short (4- to 5-residue) amino acid chain, containing L-alanine, D-glutamic acid, meso-diaminopimelic acid, and D-alanine in the case of *Escherichia coli* (a Gram-negative bacterium) or L-alanine, D-glutamine, L-lysine, and D-alanine with a 5-glycine interbridge between tetrapeptides in the case of *Staphylococcus aureus* (a Gram-positive bacterium). Peptidoglycan is

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one of the most important sources of D-amino acids in nature.

Bacterial Cell Wall Synthesis Continue.....

By enclosing the inner membrane, the peptidoglycan layer protects the cell from lysis caused by the turgor pressure of the cell. When the cell wall grows, it retains its shape throughout its life, so a rod shape will remain a rod shape, and a spherical shape will remain a spherical shape for life. This happens because the freshly added septal material of synthesis transforms into a hemispherical wall for the offspring cells.

Cross-linking between amino acids in different linear amino sugar chains occurs with the help of the enzyme DD-

transpeptidase and results in a 3-dimensional structure that is

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strong and rigid. The specific amino acid sequence and

PEPTIDOGLYCAN SYNTHESIS

The biosynthesis of PG can be divided into three (3) different stages.

The first stage occurs in the cytoplasm and leads to the synthesis of

the nucleotide sugar-linked precursors UDP- N -acetylmuramyl

p e n t a p e p t i d e (U D P - M u r N A c - p e n t a p e p t i d e) a n d U D P - N -
acetylglucosamine (UDP-GlcNAc).

In the second stage, which takes place at the cytoplasmic membrane,

precursor lipid intermediates are synthesized. The phospho-MurNAc-

pentapeptide moiety of UDP-MurNAc-pentapeptide is transferred to

the membrane acceptor bactoprenol, yielding lipid I [MurNAc-

(pentapeptide)-pyrophosphoryl-undecaprenol].

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PEPTIDOGLYCAN SYNTHESIS.....

Then, GlcNAc from UDP-GlcNAc is added to lipid I, yielding

lipid II [GlcNAc-β-(1,4)-MurNAc-(pentapeptide)-pyrophosphoryl

-undecaprenol], which is the substrate for the polymerization

reactions in bacteria that have directly cross-linked PG.

The use of a lipophilic molecule such as bactoprenol enables

the cell to transport hydrophilic precursors from the aqueous

environment of the cytoplasm, through the hydrophobic

membrane, and to the externally situated sites of incorporation

into the growing PG.

It has been suggested that the translocation of the lipid-linked

precursor from the cytoplasmic side to outer side of the

membrane, at a high rate that matches the rate of PG synthesis,

is catalyzed by a specific translocase or flippase.

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PEPTIDOGLYCAN SYNTHESIS.....

These proteins are members of the SEDS (shape, elongation, division,

and sporulation) family, and homologues have been found in many

bacteria that contain a cell wall but not in the wall-less Mycoplasma

genitalium or the archaeon Methanococcus janasschii

The facts that RodA and FtsW homologues are involved in cell wall

synthesis, that their genes are often associated in operons with a

specific pbp gene, and that the proteins are integral membrane

proteins generally predicted to have 10 transmembrane helices are in

accordance with the suggestion that they may channel the lipid

precursors to their cognate penicillin-binding proteins (PBPs).

The third and final stage of PG biosynthesis, which takes place at the outer side of the cytoplasmic membrane, involves the polymerization of the newly synthesized disaccharide-peptide units and incorporation into the growing PG. 3/15/2025 11

PEPTIDOGLYCAN SYNTHESIS.....

□ This is achieved mainly through the action of the so-called penicillin-binding proteins, which catalyze the transglycosylation and transpeptidation reactions responsible for the formation of the glycosidic and peptide bonds of the PG, respectively.

In the transglycosylation reaction, for the formation of glycan strands, the reducing end of the MurNAc of the nascent lipid-linked PG strand is likely transferred onto the C-4 carbon of the glucosamine residue of the lipid-linked PG precursor, with concomitant release of undecaprenyl-pyrophosphate.

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PEPTIDOGLYCAN SYNTHESIS.....

The undecaprenyl-pyrophosphate is then dephosphorylated to yield the lipid carrier bactoprenol, which becomes available for a second round of synthesis. It is still not clear how or whether the growing strand is released from the enzyme and what determines the length of the PG strand.

During transpeptidation, the D-Ala-D-Ala bond of one stem peptide is first cleaved and an enzyme-substrate intermediate is formed, with the concomitant release of the terminal D-Ala. The cleavage reaction provides the energy necessary for the transpeptidation reaction, which occurs outside the cytoplasmic membrane in the absence of energy donors such

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PEPTIDOGLYCAN SYNTHESIS.....

A second step involves the transfer of the peptidyl moiety to an acceptor. This acceptor is the non-alpha amino group of the dibasic amino acid in a second stem peptide (in bacteria with direct cross-linking) or the last amino acid of the cross bridge when it exists.

The reaction results in the formation of a new peptide bond between the penultimate D-alanine of a donor peptide and an amino group of the cross bridge of an acceptor peptide.

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Fig. 2: Building blocks and synthesis reactions of peptidoglycan

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synthesis

BIOSYNTHESIS OF COMPLEX LIPIDS

Storage lipids (triacylglycerols) are comprised of a glycerol backbone and three fatty acids which are bound to the glycerol by ester bindings.

Complex lipids are lipids which contain not only fatty acids, but other groups as well. They have a hydrophilic “head” and two hydrophobic fatty acid “tails”.

Their backbone can be glycerol or sphingosine. Complex lipids with glycerol backbones are called glycerophospholipids, while those with sphingosine backbones are called sphingolipids. Glycerophospholipids and sphingolipids can have many different attached groups, like alcohol, sugars, choline, etc. 3/15/2025 16

BIOSYNTHESIS OF COMPLEX LIPIDS.....

Phospholipids are complex lipids in which the hydrophilic head is comprised of phosphate and a special group, like choline, serine, inositol, or ethanolamine.

The backbone can be glycerol or sphingosine. Many phospholipids are important, like phosphatidylcholine, phosphatidylserine, phosphatidylinositol, and phosphatidylethanolamine. Phospholipids are major building blocks of cell membranes.

Sphingolipids are complex lipids whose backbone is sphingosine instead of glycerol. The hydrophilic head is comprised of a special group with or without phosphate. Like phospholipids, sphingolipids are components of cell membranes, more specifically the outer leaflet. 3/15/2025 17

SYNTHESIS OF COMPLEX LIPIDS

Synthesis of complex lipids occurs on the surfaces of the smooth ER and the inner mitochondrial membrane. Synthesis follows these steps:

1. Synthesis of the backbone (glycerol or sphingosine)
2. Attachment of fatty acids to the backbone by ester binding
3. Addition of hydrophilic head group
4. Alteration or exchange of the head group to yield the final product

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BIOSYNTHESIS OF COMPLEX LIPIDS.....

Glycerol 3-phosphate is an important precursor in glycerophospholipid synthesis. It can be synthesized by attaching a phosphate group to a glycerol (by glycerol kinase, only in the liver), or by reduction dihydroxyacetonephosphate, a glycolytic intermediate (by glycerol-3-phosphate dehydrogenase).

Fatty acids can be attached to the first and second carbon of glycerol 3-phosphate, to yield a so-called phosphatidic acid.

Attachment of serine, choline, ethanolamine, etc. to the phosphate head forms the final glycerophospholipid.

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BIOSYNTHESIS OF COMPLEX LIPIDS.....

The phosphatidic acid can also have its phosphate head removed by phosphatidic acid phosphatase, which yields a 1,2-diacylglycerol. Attachment of a third fatty acid to where the phosphate head used to be forms a triacylglycerol. This shows that triacylglycerol synthesis and complex lipid synthesis shares quite a few steps.

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Fig. 2: Synthesis of Triacylglycerol 3/15/2025 21

Fig. 3: Biosynthesis of phospholipids; The enzymes are numbered (1) Choline kinase, (2)

phosphocholinecytidyltransferase, (3) Phosphatidate phosphohydrolase, (4) Phosphocholine diacylglyceroltransferase, (5)

CTP- Phosphatidate cytidyltransferase, (6,) CDP-Diacylglycerol inositoltransferase.

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BIOSYNTHESIS OF COMPLEX LIPIDS.....

Synthesis of glycerol 3-phosphate: Two mechanisms are involved for the synthesis of glycerol 3-phosphate

1. In the liver, glycerol is activated by glycerol kinase. This enzyme is absent in adipose tissue.

2. In both liver and adipose tissue, glucose serves as a precursor for glycerol 3-phosphate. Dihydroxyacetone phosphate (DHAP) produced in glycolysis is reduced by glycerol 3-phosphate dehydrogenase to glycerol 3-phosphate.

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BIOSYNTHESIS OF COMPLEX LIPIDS.....

Addition of acyl groups to form TG: Glycerol 3-phosphate acyltransferase catalyses the transfer of an acyl group to produce

lysophosphatidic acid. DHAP can also accept acyl group, ultimately resulting in the formation of lysophosphatidic acid. Another acyl group is added to lysophosphatidic acid to form phosphatidic acid (1,2-diacylglycerolphosphate). The enzyme phosphatase cleaves off phosphate of phosphatidic acid to produce diacylglycerol. Incorporation of another acyl group finally results in synthesis of triacylglycerol, as shown on the figure 2 above

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The intermediates of TG synthesis Phosphatidic acid and

PHOSPHOLIPIDS SYNTHESIS

Phospholipids are a specialized group of lipids performing a variety of functions. These include the membrane structure and functions, involved in blood clotting and supply of arachidonic acid for the synthesis of prostaglandin.

Phospholipids are synthesized from phosphatidic acid and 1,2-diacylglycerol, intermediates in the production of triacylglycerol, as shown on figure 3 above.

Phospholipid synthesis occurs in the smooth endoplasmic reticulum. The outlined steps involved in phospholipids biosynthesis are as follows:

Formation of Lecithin and Cephalin: Choline and ethanolamine first get phosphorylated and then combine with CTP to form, respectively,

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PHOSPHOLIPIDS SYNTHESIS.....

Formation of Lecithin and Cephalin: Choline and ethanolamine first get phosphorylated and then combine with CTP to form, respectively, CDP-choline and CDP-ethanolamine

Phosphatidylcholine(lecithin) is synthesized when CDP-choline Combines with 1,2-diacylglycerol.

Phosphatidyl ethanolamine (cephalin) is produced when CDP-ethanolamine reacts with 1,2-diacylglycerol. Phosphatidyl ethanolamine can be converted to phosphatidyl choline on methylation.

Choline and ethanolamine, used for phospholipid synthesis, are mostly derived from the pre-existing phospholipids. Thus, the phospholipid synthesis starting with choline or ethanolamine is regarded as salvage

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pathway.

PHOSPHOLIPIDS SYNTHESIS.....

Choline and ethanolamine, used for phospholipid synthesis, are mostly derived from the pre-existing phospholipids. Thus, the phospholipid synthesis starting with choline or ethanolamine is regarded as salvage pathway.

Synthesis of phosphatidyl serine: Phosphatidyl ethanolamine can exchange its ethanolamine group with free serine to produce phosphatidylserine. The latter, on decarboxylation, gives the former. Formation of phosphatidylinositol: CDP-diacylglycerol produced from phosphatidic acid combines with inositol to form phosphatidyl inositol (PI).

This phospholipid contains arachidonic acid on carbon 2 of glycerol which serves as, a substrate for prostaglandin synthesis. Further, PI is
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important for signal transmission across membranes.

PHOSPHOLIPIDS SYNTHESIS.....

Synthesis of phosphatidyl glycerol and cardiolipin: CD-diacylglycerol combines with glycerol 3-phosphate to form phosphatidyl glycerol 3-phosphate, which then forms phosphatidylglycerol. The latter combines with another molecule of phosphatidylglycerol to finally produce cardiolipin.

Cardiolipin is the only phospholipid possessing antigenic properties

Formation of plasmalogens: These are phospholipids with fatty acid at carbon 1 bound by an ether linkage instead of
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ester linkage. An important plasmalogen, 1-alkenyl 2-acetyl

PHOSPHOLIPIDS SYNTHESIS.....

Formation of plasmalogens: These are phospholipids with fatty acid at carbon 1 bound by an ether linkage instead of ester linkage. An important plasmalogen, 1-alkenyl 2-acetyl glycerol 3 phosphocholine, causes blood platelet aggregation and is referred to as platelet activating factor (PAF).

Synthesis of sphingomyelins: These are phospholipids containing a complex amino alcohol, sphingosine, instead of glycerol. Palmitoyl CoA and serine combine and undergo a sequence of reaction to produce sphingosine which is then acylated to produce ceramide. Sphingomyelin is synthesized when ceramide combine with CDP-choline

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COMPLEX LIPIDS IN CELL SIGNALING

Complex lipids are found in the cell membrane. Some of the complex

lipids of the cell membrane are comprised of molecules which act as second messengers in cell signalling. When the cell gets a signal, enzymes can cleave off these second messengers from the rest of the complex lipid. These second messengers will then go on to transmit the signal inside the cell.

Phospholipids are broken down by phospholipases, of which there are four types; phospholipase A1, A2, C, and D. The different phospholipases cleave the phospholipids at different sites, as seen below. Depending on where the phospholipid is split, different parts are released.

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Fig. 3: Degradation of phospholipids by phospholipases

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FUNCTIONS OF PHOSPHOLIPASES

Phospholipase A1 specifically cleaves the fatty acid at C1 position of phospholipid resulting in lysophospholipids. The latter can be further acted by lysophospholipase, Phospholipase B act to remove the second acyl group at C2 position.

Phospholipase A2 hydrolyses phospholipids. Snake venom and bee venom are rich sources of phospholipase A2. This enzyme is found in many tissues and pancreatic juice.

Phospholipase A2 acts on phosphatidyl inositol to liberate arachidonic acid, the substrate for the synthesis of

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prostaglandins.

FUNCTIONS OF PHOSPHOLIPASES.....

Phospholipase C specifically cleaves the bond between phosphate and glycerol of phospholipids. This enzyme is present in lysosomes of hepatocytes.

Phospholipase D hydrolyses and removes the nitrogenous base from phospholipids.

The degraded products of phospholipids enter the metabolic pool and are utilized for various purposes.

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ARACHIDONIC ACID METABOLISM

Arachidonic acid (arachidonate) is a 20 carbon long polyunsaturated fatty acid. It has four double bonds. Arachidonic acid is very important, it is the precursor for prostaglandins, thromboxanes, and leukotrienes, which are very important groups of molecules in

inflammation and disease.

Arachidonic acid has double bonds beyond the 10th carbon, so it cannot be synthesized from palmitate in humans. However, plants can put double bonds anywhere they want. Humans can synthesize arachidonic acid from linoleate, an 18 carbon long polyunsaturated fatty acid with two double bonds.

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ARACHIDONIC ACID METABOLISM

Linoleate is synthesized in plants and is one of the essential fatty acids, because the animal body requires it as a substrate for arachidonic acid synthesis.

Both linoleate and arachidonic acid are omega-6 acids, referring to the fact that they have a double bond at the 6th carbon from the end.

Cells "store" arachidonic acid as one of the fatty acids of the complex lipids which make up their cell membrane. When the cell gets a particular signal, phospholipase A2 is activated, which cleaves the arachidonic acid from the phospholipid.

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Fig. 4: Overview of biosynthesis of prostaglandins and related compounds (5 - HPETE-5-Hydroperoxy- γ -linolenic acid; PG - Prostaglandins; PGI₂ - Prostacyclin

3/15/2025 2; TXA₂-Thromboxane A₂).

ARACHIDONIC ACID METABOLISM

Arachidonic acid (5,8,11,14-eicosatetraenoic acid) is the precursor for most of the prostaglandins in humans, it occurs in the endoplasmic reticulum in the following stages, as depicted on the figure above. Release of arachidonic acid from membrane bound phospholipids by phospholipase A₂

-this reaction occurs due to a specific stimuli by hormones such as epinephrine or bradykinin.

Oxidation and cyclization of arachidonic acid to PGG₂ which is then

2

converted to PCH by a reduced glutathione dependent peroxidase.

2

PGH serves as the immediate precursor for the synthesis of number of

2

prostaglandins, including prostacyclins and thromboxane.

NOTE: cyclooxygenase is an enzyme capable of undergoing self-catalysed destruction to switch off PG synthesis

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INHIBITION OF PGS SYNTHESIS

A number of structurally unrelated compounds can inhibit prostaglandin synthesis. Corticosteroids (e.g cortisol) prevent the formation of arachidonic acid by inhibiting the enzyme phospholipaseA2.

Many non-steroidal anti-inflammatory drugs inhibit the synthesis of prostaglandins, prostacyclins and thromboxanes. They do so by blocking the action of the enzyme cyclooxygenase

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INHIBITION OF PGS SYNTHESIS.....

Aspirin inhibits PG synthesis: Aspirin (acetylsalicylic acid) has been used since nineteenth century as an antipyretic (fever-reducing) and analgesic (pain relieving). The mechanism of action of aspirin however, was not known for a long period. It was only in 1971, John Vane discovered that aspirin inhibits the synthesis of PG from arachidonic acid. Aspirin irreversibly inhibits the enzyme cyclooxygenase

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BIOCHEMICAL ACTIONS OF PROSTAGLANDINS

Prostaglandins act as local hormones in their function. They, however, differ from the true hormones in many ways. Prostaglandins are produced in almost all the tissues in contrast to hormonal synthesis which occurs in specialized glands.

PGs are not stored and they are degraded to inactive products at the site of their production. Further, PCs are produced in very small amounts and have low half-lives.

Prostaglandins are involved in a variety of biological functions. The actions of PGs differ in different tissues. Sometimes, PGs bring about opposing actions in the same tissue.

Overproduction of PGs results in many symptoms which include pain, fever, nausea, vomiting and inflammation. 3/15/2025 40

BIOCHEMICAL ACTIONS OF PROSTAGLANDINS

Regulation of blood pressure : The prostaglandins (PGE, PGA and PGI 2) are vasodilator in function. This results in increased blood flow and decreased peripheral resistance to lower the blood pressure. PGs serve as agents in the treatment of hypertension.

Inflammation : The prostaglandins PGE1 and PGE2 induce the symptoms of inflammation (redness, swelling, edema etc.) due to arteriolar Vasodilation. This led to the belief that PGs are natural mediators of inflammatory reactions

of rheumatoid arthritis (involving joints), psoriasis (skin), conjunctivitis (eyes) etc. Corticosteroids are frequently used to treat these inflammatory reactions, since they

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BIOCHEMICAL ACTIONS OF PROSTAGLANDINS

Reproduction: Prostaglandins have widespread applications in the field of reproduction. PGE₂ and PGF₂ are used for the medical termination of pregnancy and induction of labor. Prostaglandins are administered to cattle to induce estrus and achieve better rate of fertilization.

Pain and fever : It is believed that pyrogens (fever producing agents) promote prostaglandin biosynthesis leading to the formation of PGE₂ in the hypothalamus, the site of regulation of body temperature. PGE₂ along with histamine and bradykinin cause pain. Migraine is also due to PGE₂. Aspirin and other non-steroidal drugs inhibit PG synthesis and thus

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